

ECOTEC_Online Student Risk Dataset

Preprocessing and Reproducibility Description

This document describes the preprocessing workflows, validation procedures, anonymization steps, and reproducibility resources associated with the ECOTEC_Online Student Risk Dataset.

1. Data Acquisition

Institutional records were extracted from academic management systems associated with online higher education programs at Universidad ECOTEC, Ecuador. The dataset integrates demographic, academic, financial, admission, and institutional variables associated with student risk analytics.

2. Initial Validation

The preprocessing workflow includes automated validation of required variables, target labels, and dataset structure. Validation scripts verify the presence of expected columns, consistency of binary target labels, and file integrity.

3. Data Cleaning

Cleaning procedures include removal of duplicated records, normalization of variable names, exclusion of rows with missing target labels, and formatting harmonization for reproducible analytical workflows.

4. Data Anonymization

Personally identifiable information was removed before public dissemination. The anonymization process excluded names, identification numbers, email addresses, telephone numbers, and direct institutional identifiers. Additional transformations and categorical aggregations were applied to reduce re-identification risk while preserving analytical utility.

5. Feature Engineering and Encoding

Benchmark scripts implement preprocessing pipelines using scikit-learn transformers. Categorical variables are encoded using OneHotEncoder, while numerical variables are processed through imputation and scaling workflows when required by the benchmark models.

6. Benchmark Reproducibility Pipelines

The repository includes reproducible benchmark pipelines implemented in Python using Random Forest and Logistic Regression classifiers. The scripts use fixed random seeds, stratified train-test splitting, and standardized preprocessing workflows to facilitate reproducibility.

7. Software Environment

The preprocessing and benchmarking workflows were implemented using Python 3.11 with the following libraries: pandas, NumPy, scikit-learn, matplotlib, openpyxl, and joblib. Package dependencies are documented in the

requirements.txt file included in the repository.

8. FAIR and Open Science Considerations

The repository was organized according to FAIR data principles to facilitate findability, accessibility, interoperability, and reuse of educational AI resources. The repository includes metadata documentation, preprocessing scripts, reproducibility resources, and citation files intended to support open-science dissemination.

9. Repository Components

The repository includes:

- Raw dataset
- Processed dataset
- Anonymized dataset
- Variable dictionary
- Codebook
- Benchmark scripts
- Figure generation scripts
- Reproducibility documentation
- FAIR/Open-science metadata resources